

CCPS616 Lab 3 – Greed is Good?

Preamble

In this lab you will write a C++ program that implements two greedy algorithms. Your implementation must be from the ground up without linking any external libraries. Your lab should be completed in a single cpp file called **greedy.cpp**, and should compile at the command line very simply as follows:

```
g++ -o lab3 greedy.cpp lab3.cpp
```

Lab Description

Part I) Cyclic Process Scheduling

The first algorithm you will implement is based on the process scheduling example from lecture 6. Your algorithm will accept a list of processes with known run times that will be executed across N CPUs. It will return the *total wait time* for all processes. This one's pretty simple, check out the provided test code to see how it should run.

Part II) Hotel Booking

In this part of the lab you will implement a greedy algorithm for scheduling hotel bookings. You will write a function that accepts as input a series of hotel room requests and the number of rooms that are available. Your function will determine a scheduling that maximizes the number of bookings that are accommodated and return those bookings. At the pseudocode level, the input will include an integer representing the number of rooms and an array of integer pairs where each pair represents a start time and end time:

```
3      // Number of rooms
      [ (1, 2), (3, 6), (5, 8), (6, 7), (6, 8) ]
```

The output of your function will be similar. It will be a 2D array of interval arrays, one array of intervals for each room. This will represent the scheduling that maximizes the number of intervals that are accommodated for a given number of rooms.

```
Room 1: [ (1, 2), (3, 6), (6, 7) ]
Room 2: [ (5, 8) ]
Room 3: [ (6, 8) ]
```

If the bookings cannot all be accommodated, your output will also include a list of those intervals that were not used in the solution. If the intervals in the above example were the same, but we only have two rooms, the output would look as follows:

```
Room 1:    [ (1, 2), (3, 6), (6, 7) ]
Room 2:    [ (5, 8) ]
No Room:   [ (6, 8) ]
```

Testing & Results

The provided lab3.cpp includes a main() function and several helper functions for generating random test data for both questions, as well as testing a couple small hardcoded examples. The style is similar to previous labs, and you should fill in the function signatures provided in greepy.cpp.

Submission

Submit your source file (**greedy.cpp**) on D2L. This lab must be submitted individually.